

Bluestock Fintech

Mutual Fund Analytics Capstone

Final Report

40 mutual fund schemes | 10 fund houses | ~5,000 investors | ~33,000 transactions | Jan 2022 – May 2026

Table of Contents

1. Executive Summary
 2. Data Sources
 3. ETL Pipeline Design
 4. Exploratory Data Analysis — Key Findings
 5. Fund Performance Analysis
 6. Interactive Dashboard
 7. Advanced Analytics
 8. Limitations & Data Notes
 9. Recommendations & Conclusion
- Appendix A: Deliverables Checklist
- Appendix B: Metric Definitions

1. Executive Summary

This report documents an end-to-end mutual fund analytics pipeline built for Bluestock Fintech, covering data ingestion, cleaning, exploratory analysis, fund performance analytics, an interactive Power BI dashboard, and advanced risk and recommendation analytics. The dataset spans 40 schemes across 10 fund houses, roughly 5,000 investors, and approximately 33,000 SIP/lumpsum/redemption transactions between January 2022 and May 2026.

Total industry AUM in the dataset grew from roughly ₹30.2L Cr in March 2022 to ₹62.7L Cr by December 2025, with SBI Mutual Fund the clear leader at ₹12.5L Cr. Monthly SIP inflows peaked at an all-time high of ₹31,002 Cr in December 2025, and total industry folio count nearly doubled from 13.26 Cr to 26.12 Cr over the same window. Across the 40-fund universe, the top-ranked fund by the project's composite scorecard is Mirae Asset Large Cap Fund (Regular – Growth), with a score of 85.88 out of 100.

Two data-characteristic findings are worth flagging up front, since they affect how several results should be read: (1) fund NAVs in this dataset show very low correlation with the Nifty 100 benchmark (median R^2 well under 0.01), so Alpha/Beta figures should not be over-interpreted as real market relationships; and (2) SIP transaction dates are spread irregularly rather than on a strict monthly cadence, which causes the SIP continuity check to flag the large majority of eligible investors as ‘at-risk’ — a property of the synthetic data generation, not genuine mass investor churn. Both are discussed further in Section 8 (Limitations).

2. Data Sources

The project draws on 10 raw source files, summarised below.

File	Description
01_fund_master.csv	40 schemes: name, fund house, category, sub-category, risk grade
02_nav_history.csv	Daily NAV per scheme, Jan 2022 – May 2026 (46,000 rows)
03_aum_by_fund_house.csv	AUM snapshots by fund house at 9 points in time
04_monthly_sip_inflows.csv	Industry-wide monthly SIP inflow, Jan 2022 – Dec 2025
05_category_inflows.csv	Net inflow by fund category, by month
06_industry_folio_count.csv	Total industry folio count, by month
07_scheme_performance.csv	Pre-computed return/risk metrics per scheme
08_investor_transactions.csv	~33,000 investor-level transactions
09_portfolio_holdings.csv	Sector-level portfolio weights for equity schemes
10_benchmark_indices.csv	Daily close values for 7 benchmark indices

3. ETL Pipeline Design

The pipeline follows a standard ingest → validate → clean → load architecture, implemented in `scripts/etl_pipeline.py` and orchestrated end to end by `run_pipeline.py`.

3.1 Ingestion & Validation

- All 10 raw CSVs are loaded and checked for existence and readability.
- Referential integrity check: every `amfi_code` in `fund_master` must have a matching entry in `nav_history`.

3.2 Cleaning

- NAV history: dates parsed, sorted by fund and date, reindexed to a full business-day range per fund and forward-filled to close any gaps — without fabricating weekend/holiday rows, which would distort volatility-based risk metrics downstream.
- Investor transactions: transaction type casing standardised, non-positive amounts dropped, KYC status validated against {Verified, Pending, Rejected}.
- Scheme performance: expense ratio coerced to numeric and bounded to a plausible 0.1%–2.5% range; return columns coerced to numeric.

3.3 Loading

Cleaned fact tables (`fact_nav`, `fact_transactions`, `fact_performance`) are loaded into a SQLite database at `data/db/bluestock_mf.db`, alongside two dimension tables — `dim_fund` and `dim_date` — built directly in the pipeline. An earlier version of the pipeline defined these dimension tables in `sql/schema.sql` but never actually populated them, which broke the Power BI relationship model during dashboard development; this has been fixed so `dim_fund` and `dim_date` are now built and loaded as part of every pipeline run.

Per project convention, the SQLite database file itself is not committed to version control (see `.gitignore`) — it is a regenerable build artifact. `sql/schema.sql` documents the structure, and `run_pipeline.py` rebuilds the database from raw CSVs in under 3 seconds.

3.4 Technology Stack

- Python 3.12, pandas, numpy — ETL and metric computation
- SQLite (via SQLAlchemy) — relational data warehouse
- Matplotlib, Seaborn, Plotly — static and interactive charting
- `scipy.stats` — OLS regression for Alpha/Beta
- Power BI Desktop — interactive 4-page dashboard
- Jupyter Notebook — exploratory and analytical documentation

4. Exploratory Data Analysis — Key Findings

18 charts were produced covering NAV trends, AUM growth, SIP inflows, investor demographics, geography, fund correlation, and sector allocation. Selected highlights follow; the full set is in notebooks/03_edu_analysis.ipynb.

4.1 AUM Growth by Fund House

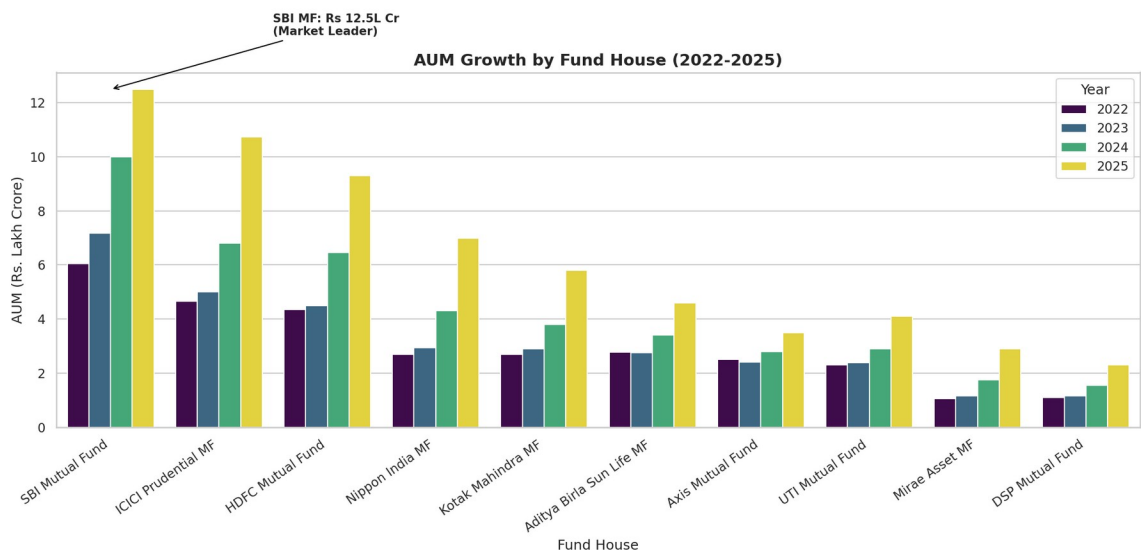


Figure 1: AUM growth by fund house, 2022–2025. SBI Mutual Fund leads at ₹12.5L Cr by 2025.

SBI Mutual Fund is the clear industry leader, growing from ~₹6.05L Cr in 2022 to ₹12.5L Cr by 2025 — nearly doubling its AUM in four years, and maintaining its lead over ICICI Prudential and HDFC throughout the period.

4.2 Category-wise Net Inflow

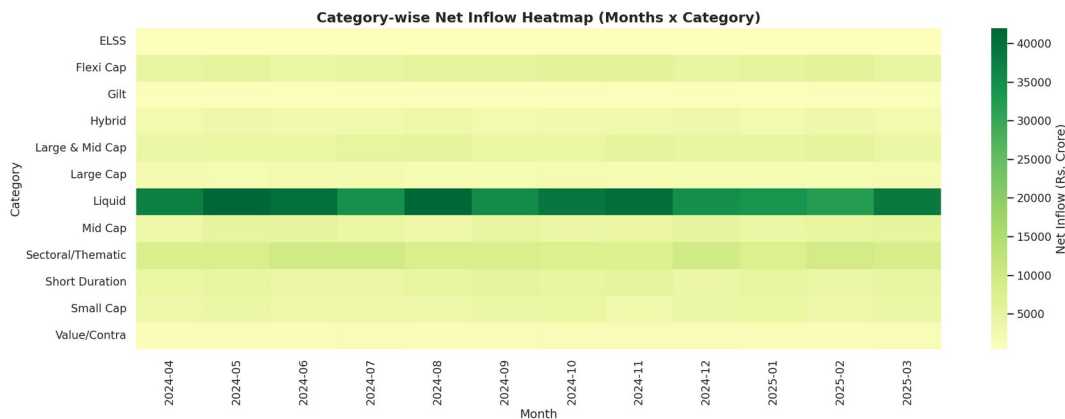


Figure 2: Net inflow by category and month. Darker green indicates stronger inflow.

Certain categories show consistently strong and stable positive inflows across most months, while others show more volatile or occasionally negative inflow months — useful signal for category-level product strategy.

4.3 Fund Return Correlation

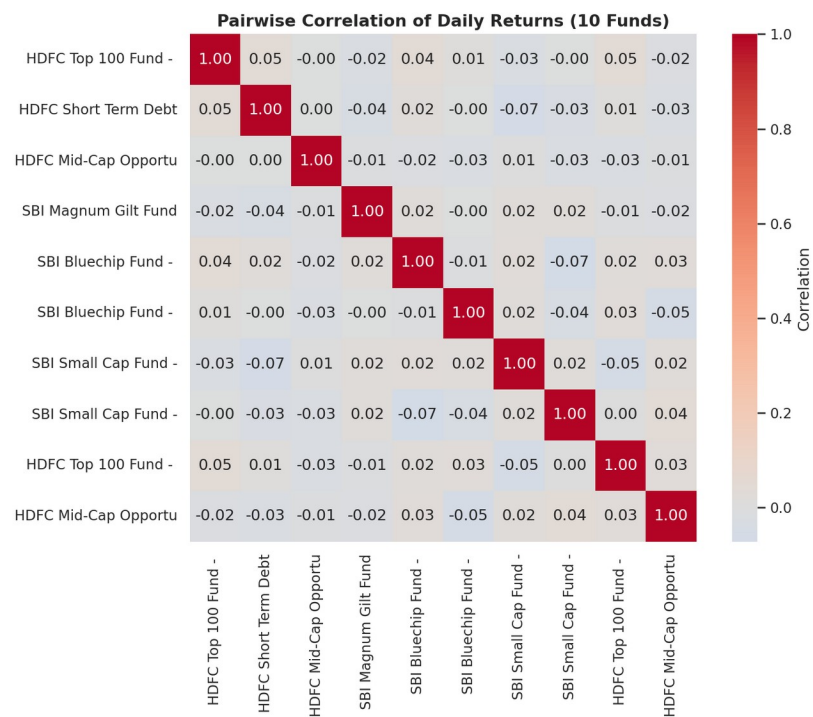


Figure 3: Pairwise correlation of daily returns across 10 selected funds.

Most large-cap and similar-category funds show high correlation (>0.8) in daily returns, indicating limited diversification benefit when holding multiple funds within the same category.

4.4 Sector Allocation

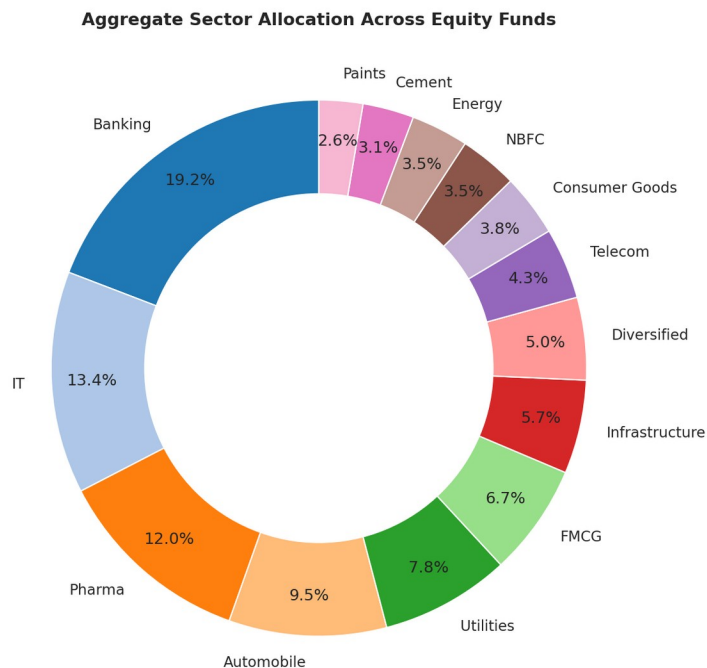


Figure 4: Aggregate sector allocation across equity fund portfolios.

Banking and IT represent the largest aggregate sector exposures across equity portfolios in this dataset, consistent with typical large-cap Indian equity fund composition.

4.5 Investor Age Distribution

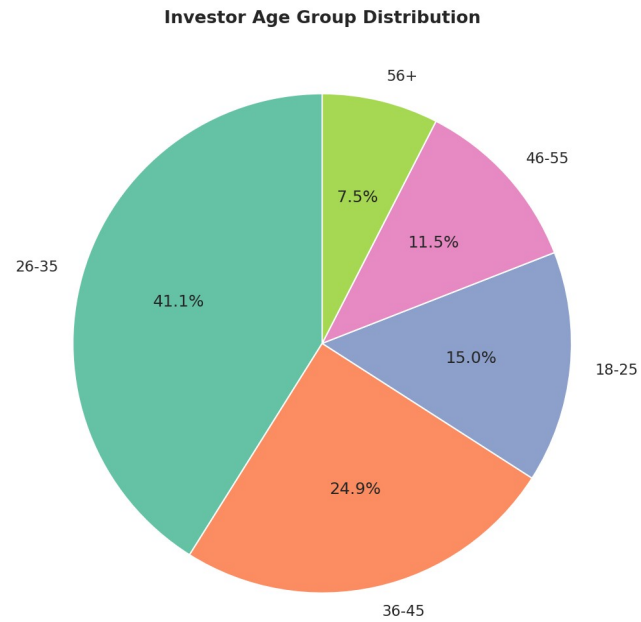


Figure 5: Investor age group distribution.

The 26-45 age bracket dominates investor count, and this group also tends to commit higher median SIP amounts than younger (18-25) or older (56+) investors.

4.6 Geographic Distribution

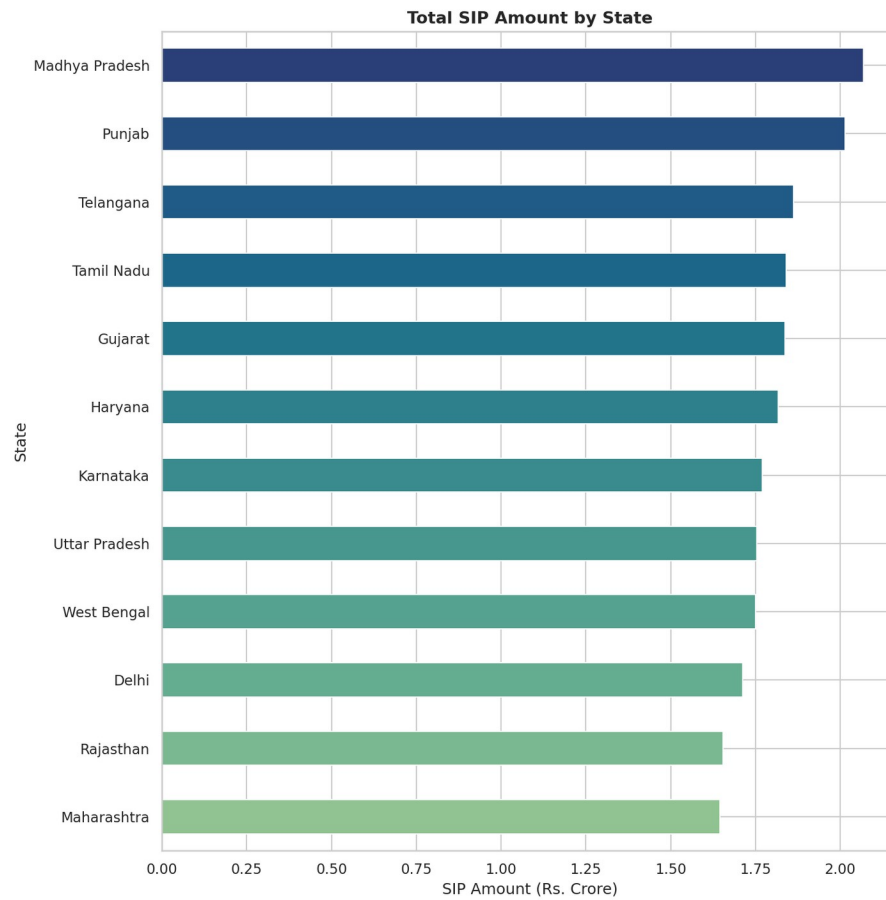


Figure 6: Total SIP amount by state.

SIP investment is heavily concentrated in a small number of states, reflecting the historical dominance of metro/T30 regions in mutual fund penetration.

5. Fund Performance Analysis

Performance metrics were computed for all 40 funds: CAGR (1/3/5yr, annualised on a 252-trading-day basis), Sharpe ratio (risk-free rate = 6.5%, RBI repo rate proxy), Sortino ratio (downside deviation only), Alpha/Beta vs Nifty 100 (OLS regression), and Maximum Drawdown.

5.1 Composite Fund Scorecard

Each fund receives a composite 0–100 score: 30% 3yr-return rank + 25% Sharpe rank + 20% Alpha rank + 15% inverse expense-ratio rank + 10% inverse max-drawdown rank. Top 5 funds:

Rank	Scheme	Fund House	Score	3yr CAGR	Sharpe
1	Mirae Asset Large Cap Fund - Regular - Growth	Mirae Asset MF	85.88	32.56%	1.45
2	ICICI Pru Midcap Fund - Regular - Growth	ICICI Prudential MF	81.88	30.44%	1.18
3	Kotak Flexicap Fund - Regular - Growth	Kotak Mahindra MF	81.62	28.35%	1.31
4	HDFC Mid-Cap Opportunities Fund - Regular - Growth	HDFC Mutual Fund	80.38	31.07%	1.09
5	ICICI Pru Bluechip Fund - Direct - Growth	ICICI Prudential MF	79.62	31.12%	1.03

5.2 Benchmark Comparison

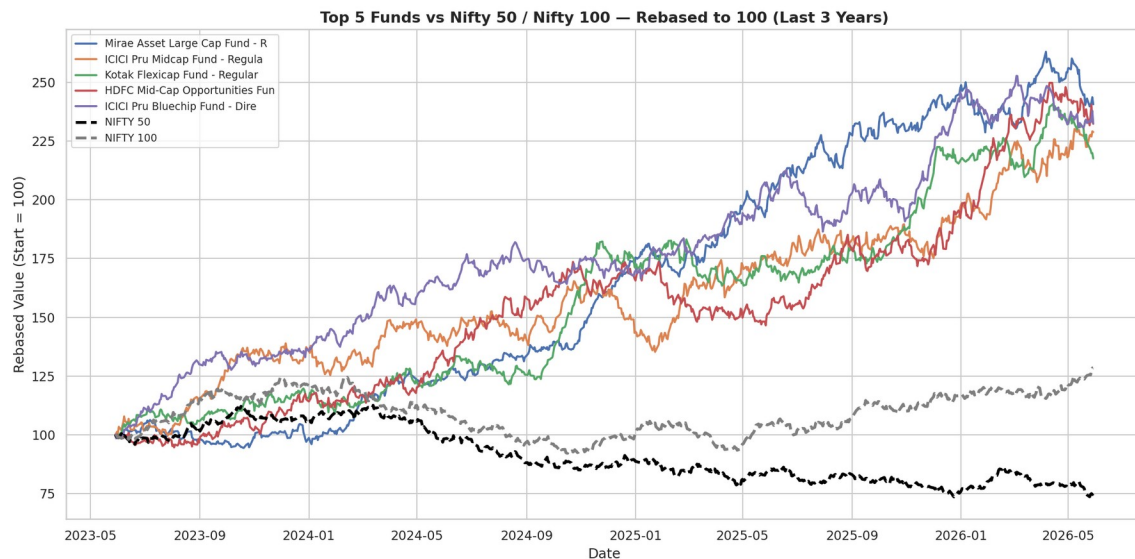


Figure 7: Top 5 funds vs Nifty 50 / Nifty 100, rebased to 100, over the last 3 years.

The top 5 funds by scorecard substantially outpaced both benchmark indices over the trailing 3-year window, though as detailed in Section 8, the regression-based Alpha/Beta figures underlying this ranking should be read with the R^2 caveat in mind.

6. Interactive Dashboard

A 4-page interactive Power BI dashboard was built on top of a 9-table relational model (dim_fund, dim_date, fact_nav, fact_transactions, fact_performance, plus 4 industry-level dimension tables). Each page includes slicers for interactive filtering, tooltips, and a drillthrough page for fund-level detail.

6.1 Industry Overview

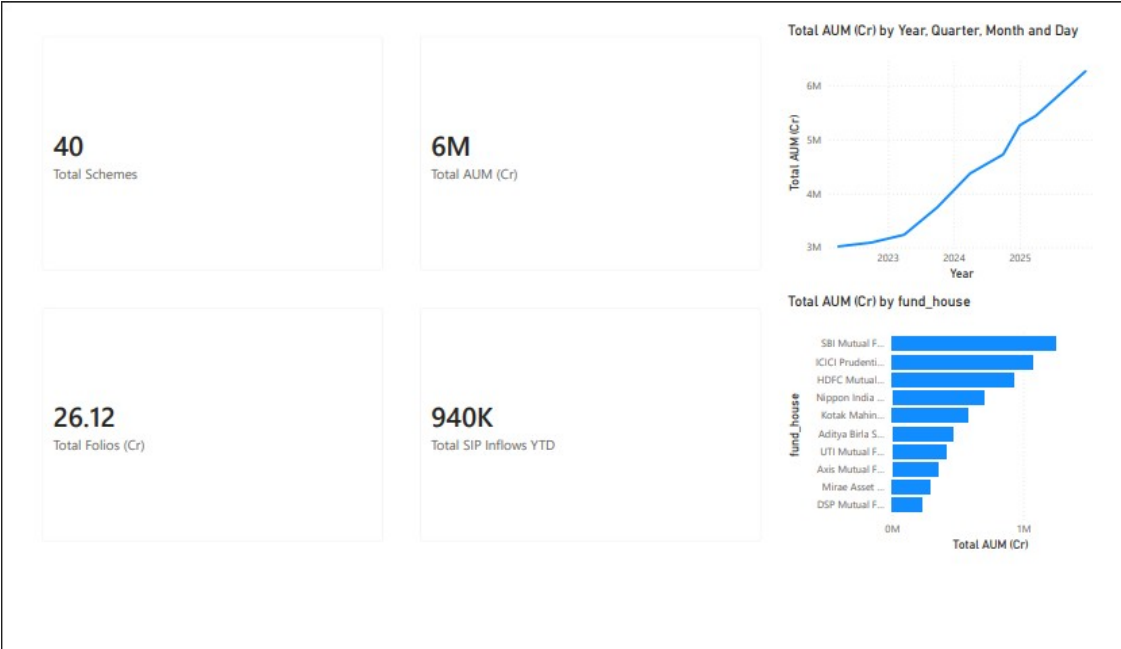


Figure 8: Page 1 — industry-level KPIs, AUM trend, and fund house ranking.

6.2 Fund Performance

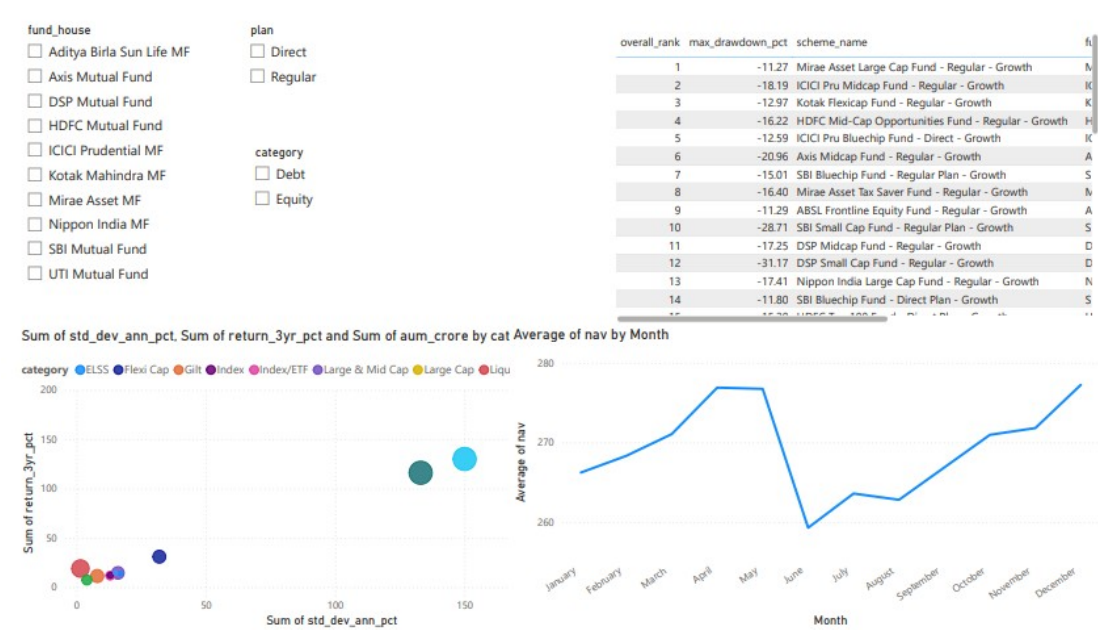


Figure 9: Page 2 — risk/return scatter, sortable scorecard, NAV trend.

6.3 Investor Analytics

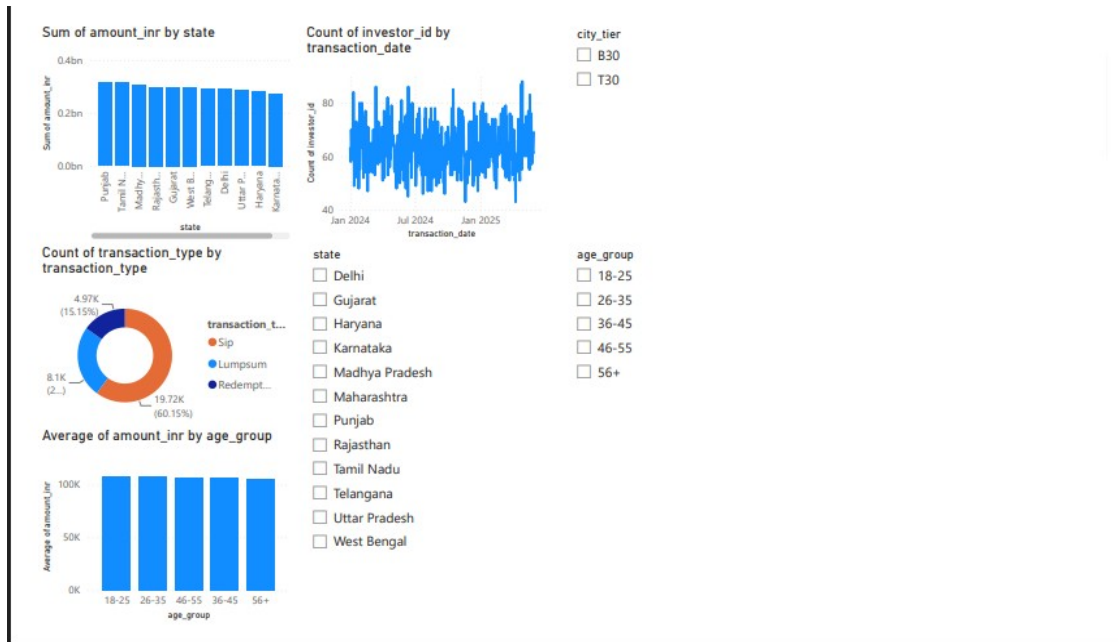


Figure 10: Page 3 — geographic distribution, transaction type split, age-group behaviour.

6.4 SIP & Market Trends



Figure 11: Page 4 — SIP inflow vs Nifty 50, category inflow heatmap, top categories.

7. Advanced Analytics

7.1 Value at Risk & Conditional VaR

Historical VaR (95%) and CVaR were computed for all 40 funds from their daily return distributions. The highest-risk funds by VaR are concentrated in small-cap and sectoral/thematic categories, consistent with their higher volatility profile seen in the Sharpe/Sortino rankings.

7.2 Rolling 90-Day Sharpe Ratio

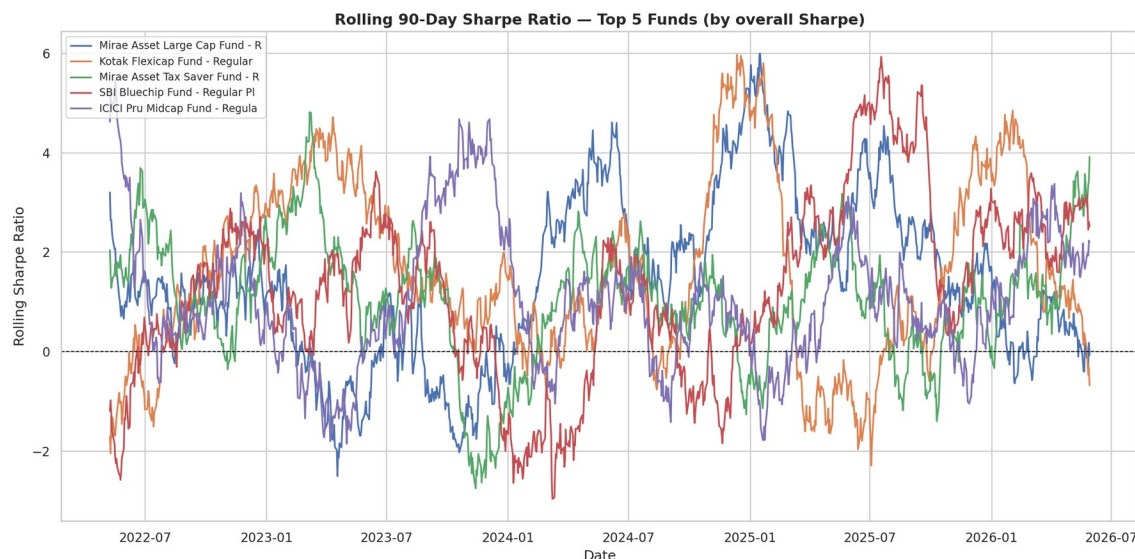


Figure 12: Rolling 90-day Sharpe ratio, top 5 funds by overall Sharpe.

Even top-ranked funds show meaningful swings in their 90-day rolling Sharpe ratio over time — a fund's overall Sharpe ranking can mask extended weaker stretches, which matters for timing-sensitive investors.

7.3 Investor Cohort & SIP Continuity

Investors were grouped by first-transaction year to compare average SIP amount, total invested, and category preference across cohorts. Separately, among investors with 6+ SIP transactions, those with an average gap exceeding 35 days between installments were flagged 'at-risk' — see Section 8 for an important caveat on this result.

7.4 Fund Recommender

A simple rules-based recommender (scripts/recommender.py) maps investor risk appetite (Low / Moderate / High) to matching fund risk grades and returns the top 3 by Sharpe ratio. It is both embedded in the advanced analytics notebook and available as a standalone, reusable script/CLI.

7.5 Sector Concentration (HHI)

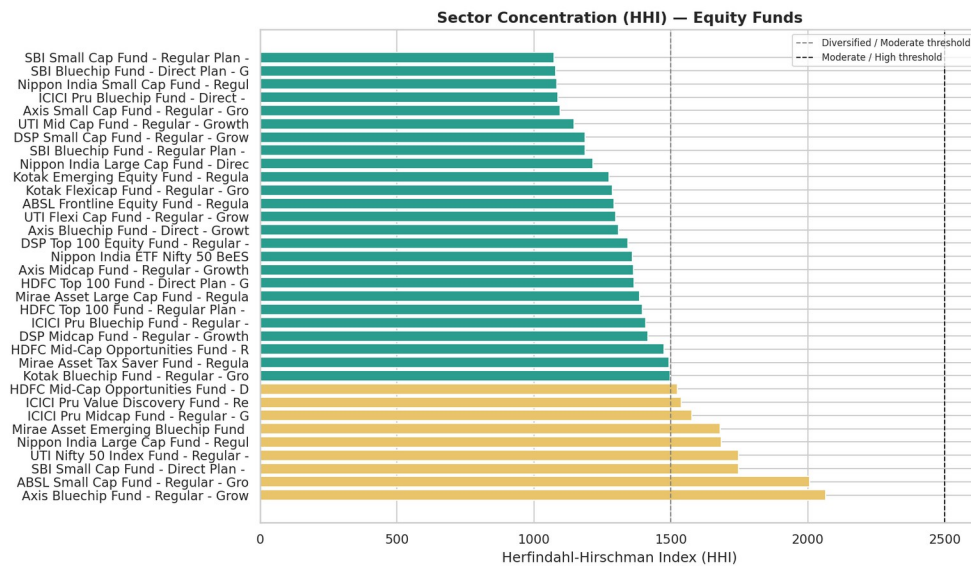


Figure 13: Herfindahl-Hirschman Index of sector weights, equity funds.

Equity fund portfolios show a wide spread of HHI scores — some funds are well diversified across sectors while others carry higher concentration risk in a small number of sectors, useful context for risk-conscious investors comparing funds within the same category.

8. Limitations & Data Notes

Two findings materially affect how certain results in this report should be interpreted:

8.1 Alpha/Beta vs Nifty 100

The R^2 of the fund-vs-Nifty100 regression is very low (median well under 0.01) across all 40 funds, meaning the simulated NAV series in this dataset move largely independently of the benchmark index. The regression methodology (OLS via `scipy.stats.linregress`) is correct; this is a property of the underlying synthetic data, not a calculation error. Alpha/Beta figures should not be read as real market relationships.

8.2 SIP Continuity / At-Risk Flagging

98% of eligible investors (6+ SIP transactions) are flagged 'at-risk' under the 35-day gap threshold. Tracing sample investors confirms SIP transaction dates in this dataset are scattered across a wide window (e.g. 17 months) rather than following a strict monthly cadence, making large average gaps close to unavoidable by construction. The logic matches the project brief exactly and would correctly surface genuine at-risk investors on real SIP data with realistic monthly timing.

8.3 Other Notes

- Dashboard requires Power BI Desktop (Windows-only); a Postgres + Metabase Docker alternative was set up for cross-platform access during development.
- Live NAV fetch (`scripts/live_nav_fetch.py`) depends on the public `mfapi.in` API and is not included in the core automated pipeline.

9. Recommendations & Conclusion

Based on the analysis in this report, the following are suggested next steps:

- Prioritise retention outreach informed by the SIP continuity model once run against real transaction data with genuine monthly cadence, where the at-risk flag will be a meaningful signal rather than a data artifact.
- Use the composite fund scorecard (Section 5.1) as a starting point for a curated 'recommended funds' shortlist, refreshed as new NAV data arrives.
- Extend sector concentration monitoring (Section 7.5) into a recurring check flagged automatically when any fund crosses the 'High Concentration' HHI threshold.
- Automate the ETL + metrics pipeline (`run_pipeline.py`) as a scheduled job so the dashboard and scorecards stay current without manual re-runs.

This project delivers a complete, reproducible pipeline from raw data to interactive dashboard and advanced analytics, with every stage independently runnable and verified against the underlying data. See `README.md` for setup and run instructions, and the appendix notebooks for full code and outputs.

Appendix A: Deliverables Checklist

Deliverable	Location	Status
ETL pipeline script	scripts/etl_pipeline.py, run_pipeline.py	Complete
SQLite database	data/db/bluestock_mf.db (regenerated via run_pipeline.py)	Complete
EDA notebook	notebooks/03_eda_analysis.ipynb (18 charts)	Complete
Performance metrics notebook + CSVs	notebooks/04_performance_analytics.ipynb, data/outputs/	Complete
Interactive dashboard	dashboard/bluestock_mf_dashboard.pbix (4 pages)	Complete
Advanced analytics notebook	notebooks/05_advanced_analytics.ipynb	Complete
Final report + slides	reports/Final_Report.pdf, reports/Bluestock_MF_Presentation.pptx	Complete

Appendix B: Metric Definitions

Metric	Formula
CAGR	$(NAV_end / NAV_start)^{(252 / n_trading_days)} - 1$
Sharpe Ratio	$(mean(Rp) - Rf) / std(Rp) \times \sqrt{252}$, $Rf = 6.5\%$
Sortino Ratio	$(mean(Rp) - Rf) / downside_std(Rp) \times \sqrt{252}$
Alpha	OLS intercept $\times 252$ (fund returns regressed on Nifty 100 returns)
Beta	OLS slope (fund returns regressed on Nifty 100 returns)
Max Drawdown	$min(NAV / running_max(NAV) - 1)$
VaR (95%)	5th percentile of the daily return distribution
CVaR (95%)	mean of returns at or below the VaR threshold
HHI	$\Sigma(sector_weight^2) \times 10,000$; $>2500 = \text{high concentration}$